XIANZHEN LUO

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EDUCATION

Harbin Institute of Technology **Harbin, China**
Ph.D. student in Code Intelligence Sep 2022 – Present
Supervisor: *Prof. Wanxiang Che and Assoc. Prof. Qingfu Zhu*

- **Honors**: National Scholarship

Harbin Engineering University **Harbin, China**
Bachelor in Computer Science Sep 2018 – Jun 2022

- **Honors**: National Scholarship, 2 x ICPC Regional Silver Medals, 2 x ICPC EC-Final Bronze Medals.

PUBLICATIONS

ICLR 2026 **How Many Code and Test Cases Are Enough? Evaluating Test Cases Generation from a Binary-Matrix Perspective**
Xianzhen Luo*, *Jinyang Huang**, *Wenzhen Zheng, Qingfu Zhu, Mingzheng Xu, Yiheng Xu, Yuantao Fan, Libo Qin, Wanxiang Che*

🏆 ACL 2025 Outstanding Paper **Turning Trash into Treasure: Accelerating Inference of Large Language Models with Token Recycling**
Xianzhen Luo, *Yixuan Wang, Qingfu Zhu, Zhiming Zhang, Xuanyu Zhang, Qing Yang, Dongliang Xu*

ACL 2025 **ChartCoder: Advancing Multimodal Large Language Model for Chart-to-Code Generation**
*Xuanle Zhao**, ***Xianzhen Luo****, *Qi Shi, Chi Chen, Shuo Wang, Wanxiang Che, Zhiyuan Liu, Maosong Sun*

📢 ACL 2025 Oral **OpenCoder: The Open Cookbook for Top-Tier Code Large Language Models**
Siming Huang, Tianhao Cheng, Jason Klein Liu, Weidi Xu, JIARAN HAO, Liuyihan Song, Yang Xu, Jian Yang, Jiaheng Liu, Chenchen Zhang, Linzheng Chai, Ruifeng Yuan, ***Xianzhen Luo****, Qiufeng Wang, YuanTao Fan, Qingfu Zhu, Zhaoxiang Zhang, Yang Gao, Jie Fu, Qian Liu, Houyi Li, Ge Zhang, Yuan Qi, Xu Yinghui, Wei Chu, Zili Wang.*

ACL 2025 (Findings) **ChartEdit: How Far Are MLLMs From Automating Chart Analysis? Evaluating MLLMs' Capability via Chart Editing**
Xuanle Zhao, Xuexin Liu*, Yang Haoyue**, ***Xianzhen Luo***, *Fanhu Zeng, Jianling Li, Qi Shi, Chi Chen*

KDD 2025 **Advancing Tool-Augmented Large Language Models via Meta-Verification and Reflection Learning**
Zhiyuan Ma, Jiayu Liu, ***Xianzhen Luo****, Zhenya Huang, Qingfu Zhu, Wanxiang Che*

EMNLP 2024 **Python is Not Always the Best Choice: Embracing Multilingual Program of Thoughts**
Xianzhen Luo*, Qingfu Zhu, Zhiming Zhang, Libo Qin, Xuanyu Zhang, Qing Yang, Dongliang Xu, Wanxiang Che*

EMNLP 2024 **Make Some Noise: Unlocking Language Model Parallel Inference Capability through Noisy Training**
*Yixuan Wang**, ***Xianzhen Luo****, *Fuxuan Wei, Yijun Liu, Qingfu Zhu, Xuanyu Zhang, Qing Yang, Dongliang Xu, Wanxiang Che*

COLING 2024 **A Survey on Natural Language Processing for Programming**
Qingfu Zhu, ***Xianzhen Luo****, Fang Liu, Cuiyun Gao, Wanxiang Che*

ACL 2022 (Findings) **Inverse is Better! Fast and Accurate Prompt for Few-shot Slot Tagging**
Yutai Hou, Cheng Chen, ***Xianzhen Luo****, Bohan Li, Wanxiang Che.*

AI Open 2022 **Augmented and challenging datasets with multi-step reasoning and multi-span questions for Chinese judicial reading comprehension**
Qingye Meng, Ziyue Wang, Hang Chen, ***Xianzhen Luo****, Baoxin Wang, Zhipeng Chen, Yiming Cui, Dayong Wu, Zhigang Chen, Shijin Wang*

Arxiv **CVE-Factory: Scaling Expert-Level Agentic Tasks for Code Security Vulnerability**
Xianzhen Luo*, *Jingyuan Zhang*, Shiqi Zhou*, Rain Huang**, *Chuan Xiao, Qingfu Zhu, Zhiyuan Ma, Xing Yue, Yang Yue, Wencong Zeng, Wanxiang Che*

Arxiv **From Code Foundation Models to Agents and Applications: A Practical Guide to Code Intelligence**
Core Contributor

Arxiv **Scaling Laws for Code: A More Data-Hungry Regime**
Xianzhen Luo*, *Wenzhen Zheng**, *Qingfu Zhu, Rongyi Zhang, Houyi Li, Siming Huang, Yuantao Fan, Wanxiang Che*

Arxiv **Step-3 is Large yet Affordable: Model-system Co-design for Cost-effective Decoding**
Core Contributor

Arxiv **Success is in the Details: Evaluate and Enhance Details Sensitivity of Code LLMs through Counterfactuals**
Xianzhen Luo, *Qingfu Zhu, Zhiming Zhang, Mingzheng Xu, Tianhao Cheng, Yixuan Wang, Zheng Chu, Shijie Xuyang, Zhiyuan Ma, YuanTao Fan, Wanxiang Che.*

Arxiv **Is Compression Really Linear with Code Intelligence?**
*Shijie Xuyang**, ***Xianzhen Luo****, *Tianhao Cheng, Zheng Chu, Houyi Li, Ziqi Wang, Siming Huang, Qingfu Zhu, Qiufeng Wang, Xiangyu Zhang, Shuigeng Zhou, Wanxiang Che.*

Arxiv **Automated Snippet-Alignment Data Augmentation for Code Translation**
Zhiming Zhang, Qingfu Zhu, ***Xianzhen Luo***, *Yixuan Wang, Bohan Li, Wanxiang Che*

Arxiv **Format-Adapter: Improving Reasoning Capability of LLMs by Adapting Suitable Format**
Dingzirui Wang, Xuanliang Zhang, Rongyu Cao, Longxu Dou, ***Xianzhen Luo***, *Yingwei Ma, Qingfu Zhu, Wanxiang Che, Binhua Li, Fei Huang, Yongbin Li*

Arxiv **Semi-Instruct: Bridging Natural-Instruct and Self-Instruct for Code Large Language Models**
Xianzhen Luo, *Qingfu Zhu, Zhiming Zhang, Xu Wang, Qing Yang, Dongliang Xu, Wanxiang Che*

PROJECTS

CVE-Factory: Scaling Expert-Level Agentic Tasks for Code Security Vulnerability Oct 2025 – Present
- A fully automated framework for reproducing CVE vulnerabilities, achieving 95% alignment rate with reproduction from code security experts.
- Released 1000+ executable CVE environments, training data, and Abacus-cve model (32B) that surpasses Qwen3-Coder-480B and Claude Sonnet 4.

Huozi: An Open-Source Universal LLM | 12 200 Mar 2023 – May 2023
- Served as a core developer, responsible for the collection and curation of data for code pretraining and post-training.
- Led the collection and organization of Chinese pretraining datasets.

Abacus: A Lightweight Code LLM | 2 47 Oct 2023 – Sep 2024
- Abacus with 2.7B parameters, outperforms other Code LLMs with parameters ≤3B such as Stable Code-3B and Granite-3B-Code on both coding and general language tasks.
- Led post-training data construction.

EXPERIENCE

Kuaishou Technology **Beijing, China**
Kstar *Research Intern* Aug 2025 – Present
- Focus on terminal agent development and research.

StepFun AI. **Beijing, China**
Research Intern Dec 2024 – Jul 2025
- The code aspects of LLM pretraining.
- Developed code data cleaning, training & evaluation pipelines. Provided core code pretraining data for Step3 LLM. Implemented several specialized pretraining tasks/strategies on code.

Du Xiaoman (Beijing) Science Technology Co., Ltd. **Beijing, China**
University-Industry Collaboration Researcher Nov 2023 – Sep 2024
- Explore research to enhance the code generation ability of Code LLMs, leverage code to assist other tasks, and accelerate LLM inference.

Joint Laboratory of HIT and iFLYTEK Research (HFL), China. **Beijing, China**
Research Intern Mar 2022 – Aug 2022
- Focus on Chinese Spell Check.